# Project Guidelines **F . M = 10**

Each group must design and implement a **database system** for a bioinformatics-related problem (e.g., Gene Database, Patient Management, DNA Sequence, Laboratory Information System, Protein Database, etc.) You are free to choose any suitable dataset in your domain. Your project should include the following minimum requirements:

1. **Project Title:** your own based on your data.
2. **ER Diagram** (Entity Relationship Diagram) **(In report)**.
3. **Minimum 4 related tables**.
4. **Primary Key** and **Foreign Key** relationships between tables where applicable.
5. **Appropriate SQL Data Types** for all attributes.
6. **Appropriate Constraints**, such as:
    - NOT NULL
    - UNIQUE
    - PRIMARY KEY
    - FOREIGN KEY
    - CHECK (if applicable)
7. **Sample Data** (minimum 5 records per table).
8. Screenshots of **(In report)**:
    - Created tables
    - Inserted data
    - Query execution and results

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GRADING POLICY

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You are required to demonstrate the full implementation of your project in MySQL. During the evaluation, you should be able to explain your database design, table creation, relationships, constraints, data insertion, and SQL queries sub query. Be prepared to execute your SQL statements and answer questions related to your project.

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1. Do tables have PRIMARY KEY and FOREIGN KEY?
2. Are data types chosen appropriately?
3. Constraints added (NOT NULL, UNIQUE, CHECK)?
4. Are database tables normalized (up to an appropriate normal form)?
5. Are the table relationships designed to support SQL JOIN operations?
6. Do queries work correctly?
7. Well documented?

# Project Report Format ( at least 5 - 6  Pages)

1. **Introduction**
   - Background of the selected bioinformatics data. **(one paragraph)**
   - Objectives. (**In point form**)
2. **Database Design**
   - Design an ER diagram for your database showing relationships, cardinality, etc. You may use [draw.io](draw.io) or any other ER diagram tool of your choice.
   - ER Diagram generated by MYSQL reference: https://www.youtube.com/watch?v=BjPsm7Ny1MA
3. **Implementation**
   - Short description of tables and attributes, Sample data
   - SQL queries used (CREATE, INSERT, SELECT, UPDATE, DELETE, JOIN, Aggregate Functions, Subqueries, etc.)
4. **Results**
   - Screenshots of any query implemented
5. **Conclusion:**
   - Write a **clear and concise conclusion (approximately 5–6 lines)** summarizing your project, the database you developed, and the key outcomes. Avoid lengthy or unnecessary explanations.
6. **References** (if applicable)

***Note:*** *This is a sample report format. You are encouraged to organize and present your report with additional details or sections (if applicable), while ensuring that all the mentioned contents are covered.*

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***Hint (Example)****: Sample SQL Table Structure for a Genomics Database given below.*
*You are encouraged to develop and customize your own schema rather than copying this example directly.*

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```
- Table 1: Organisms
CREATE TABLE Organisms (
   Organism_ID INT PRIMARY KEY,
   Scientific_Name VARCHAR(100) NOT NULL UNIQUE,
   Genome_Size INT
);

-- Table 2: Chromosomes
CREATE TABLE Chromosomes (
   Chromosome_ID INT PRIMARY KEY,
   Organism_ID INT NOT NULL,
   Chromosome_Number INT NOT NULL,
   Length INT,
   FOREIGN KEY (Organism_ID) REFERENCES Organisms(Organism_ID));
```

```sql
-- Table 3: Genes
CREATE TABLE Genes (
   Gene_ID INT PRIMARY KEY,
   Gene_Symbol VARCHAR(50) NOT NULL UNIQUE,
   Chromosome_ID INT,
   Start_Position INT NOT NULL,
   End_Position INT NOT NULL,
   FOREIGN KEY (Chromosome_ID) REFERENCES Chromosomes(Chromosome_ID)
);

-- Table 4: Sequences
CREATE TABLE Sequences (
   Sequence_ID INT PRIMARY KEY,
   Gene_ID INT,
   Sequence_Data TEXT NOT NULL,
   Sequence_Length INT,
   Upload_Date DATE DEFAULT (CURRENT_DATE),
   FOREIGN KEY (Gene_ID) REFERENCES Genes(Gene_ID)
);
-- Table 5: Mutations
CREATE TABLE Mutations (
   Mutation_ID INT PRIMARY KEY,
   Gene_ID INT,
   Mutation_Type VARCHAR(50) NOT NULL,
   Position INT NOT NULL,
   Frequency DECIMAL(5,4),
   CHECK (Frequency BETWEEN 0 AND 1),
   FOREIGN KEY (Gene_ID) REFERENCES Genes(Gene_ID)
);
```